Report Deep Learning Assignment 1

Introduction:

During this assignment we had to do 4 experiments:

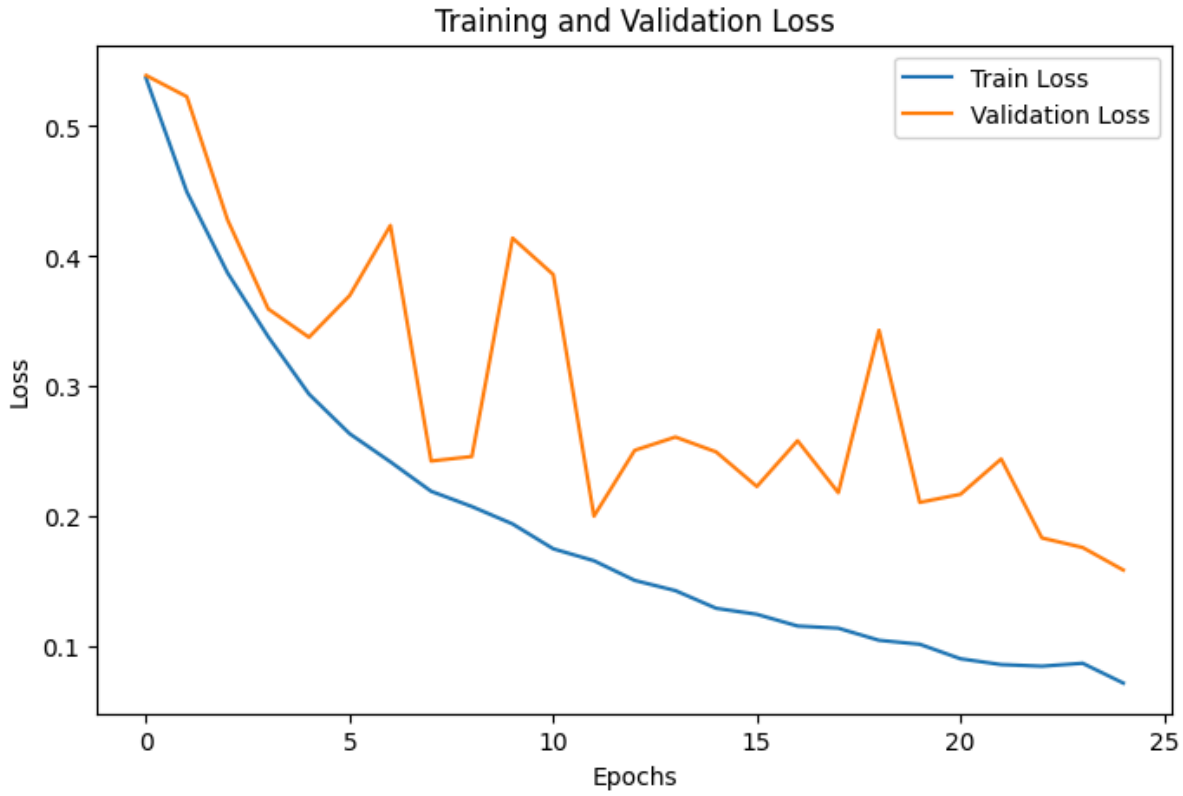
- Experiment 1:
    - Train a CNN model on the *Cats & Dogs* dataset
    - Using the same architecture, train the model from scratch on *Stanford Dogs.*
- Experiment 2: Transfer Learning using the model trained on the *Stanford Dogs* to train the model on the *Cat & Dogs* dataset for 50 Epochs, replacing the last layer (adapting to the output of the *Cat & Dogs* dataset).
- Experiment 3: Transfer Learning using the model trained on the *Stanford Dogs* to train the model on the *Cat & Dogs* dataset, replacing the last layer (adapting to the output of the *Cat & Dogs* dataset) and the 2 first convolution layers.
- Experiment 4: Transfer Learning using the model trained on the *Stanford Dogs* to train the model on the *Cat & Dogs* dataset, replacing the last layer (adapting to the output of the *Cat & Dogs* dataset) and the 2 last convolution layers.

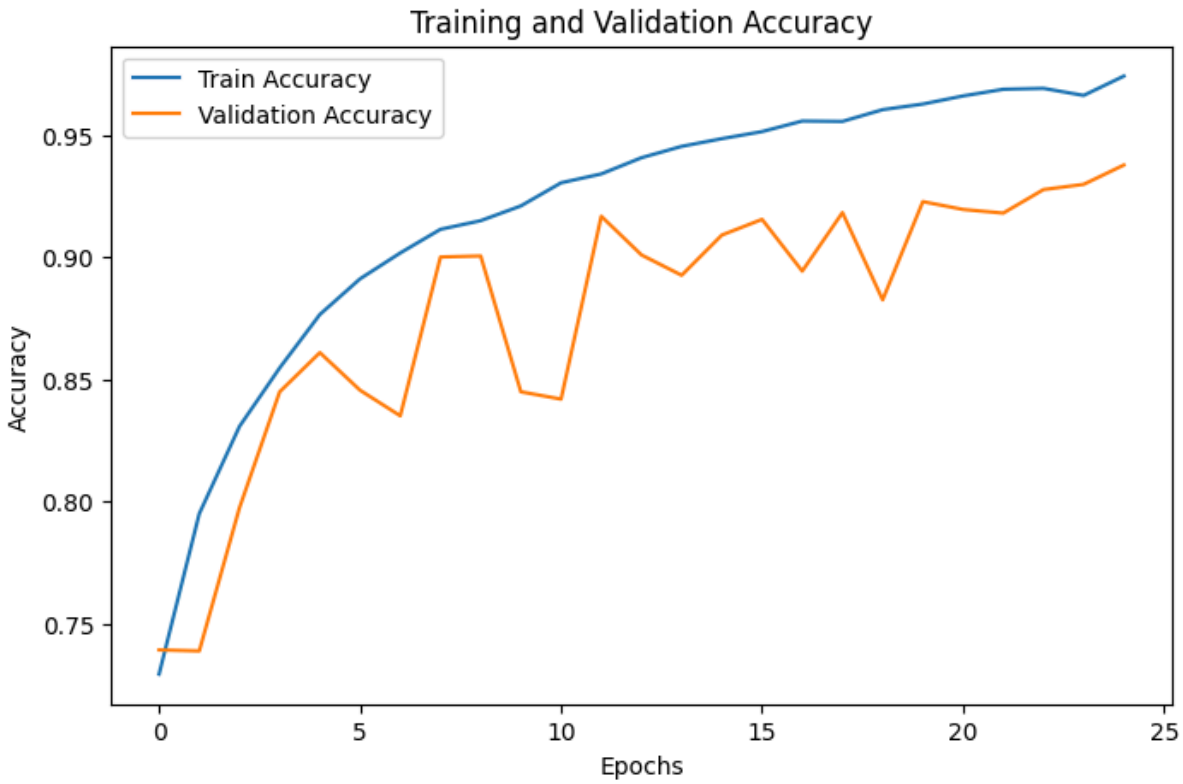
During this work, we used PyTorch to implement the architecture and do the training.

Experiment 1:

Base Model Cats & Dogs

We trained the model on the Cats & Dogs dataset using the same hyperparameters as the keras Tutorial (but adapted to Pytorch), obtaining these results:

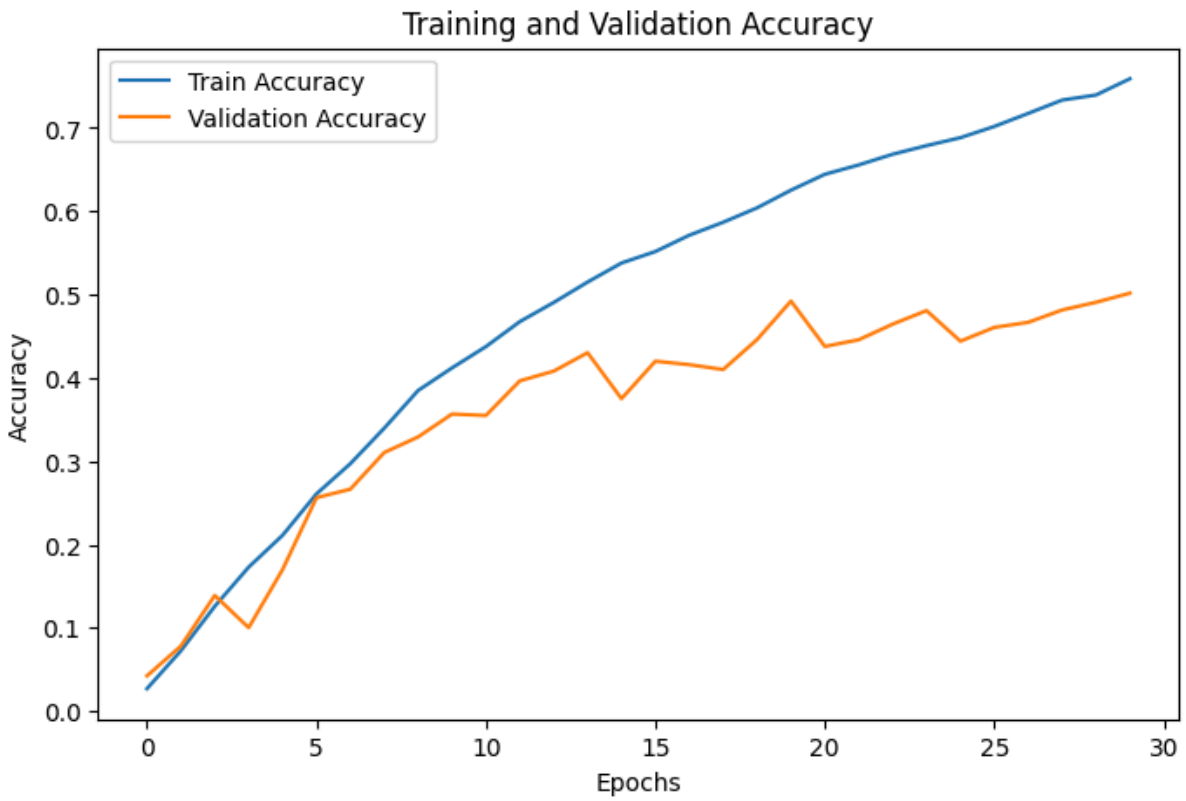




In these 2 plots you can see that in 25 Epochs, the model loss successfully converges downward and the accuracy upward, with a loss of 0.18 and an accuracy of 93% on validation set. This indicates that the model performs well on the specific task of identifying cats and dogs.
We can still note that the model had a mild overfitting during the training, the validation loss is a bit higher than the training one.

Model Stanford Dogs

We then trained a second model, with the same architecture, for 30 Epochs, changing the loss function to CrossEntropy to match the number of classes.
We also added a weight decay of 0.0001 for regularization as we saw the risk of overfitting before.

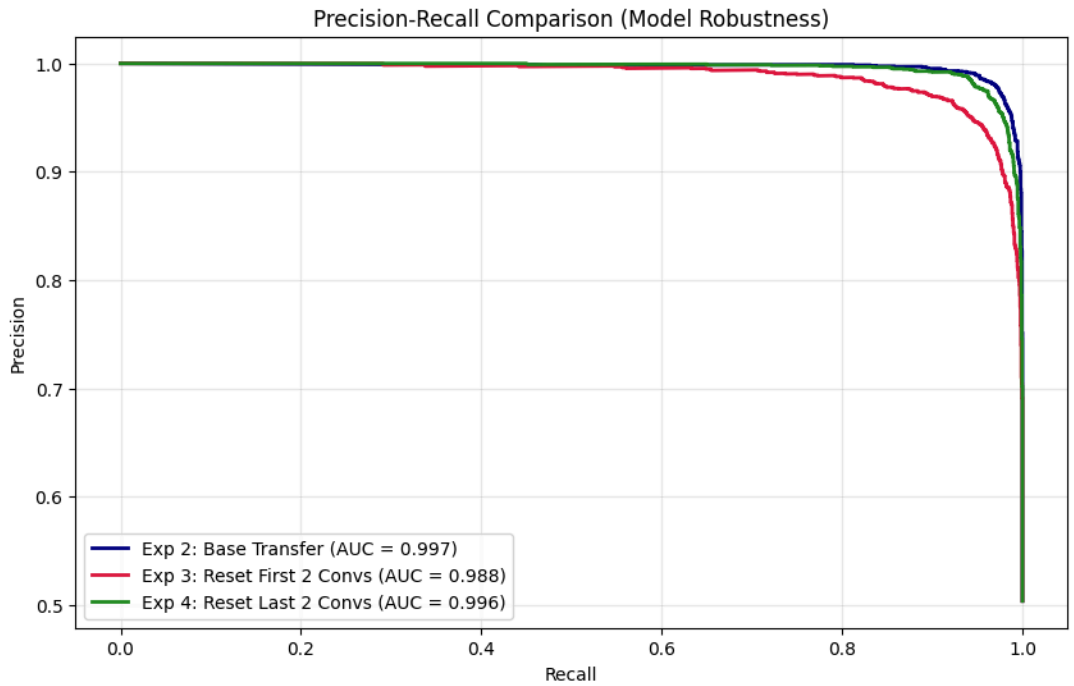


During the 30 Epochs we can see that the model performed really badly, with a loss of 2.0 and an accuracy of 0.5 on validation set.

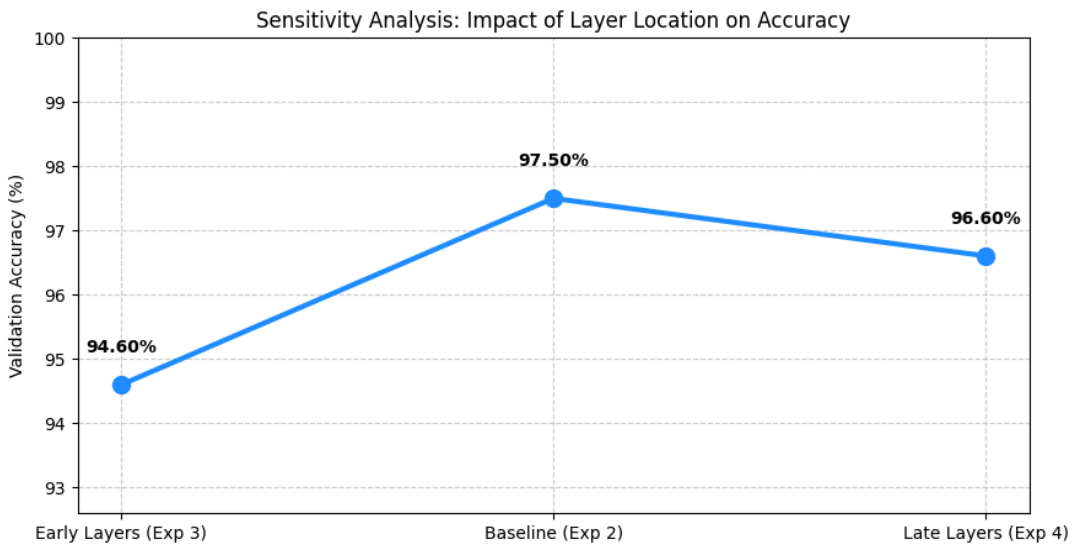
This could be mainly explained by the fact that this architecture is made for the Cats & Dogs dataset not for the Stanford Dogs dataset. Also adding more time on the training could help the model to converge to a better solution.

Comparative Study

We compared the models obtained from Experiments 2, 3 and 4 in order to see how modifications to the convolutional layers can affect the training of the models. A learning rate of 0.0001 was used for all 3 models, in order to ensure fairness among comparisons. Experiment 2 was trained on 50 epochs, while Experiment 3 and 4 were trained on 25 epochs, reasoning behind it being that upon analyzing the output for Experiment 2, after 25 epochs the improvements were minimal while the computational time was fairly big. Thus, we decided that it would be best to train the last 2 experiments on only 25 epochs.



As it can be seen from the graphs, Experiment 2 has the highest accuracy, the reason



behind it being that we simply apply transfer learning on it by using the base model, which is the Stanford Dogs model, and modifying its output layer. It reaches an accuracy of 97.5% which is satisfying considering our base model had an accuracy of ~50%.

Furthermore, looking at Experiment 3 we can see an accuracy of 94.6%. It has a slightly lower accuracy due to the fact that we replace the first two convolutional layers and the output layer, which more simply put, requires the model to re-learn primitive features such as edge detection.

Through Experiment 4, we are testing the logic of the model. The layers we replaced, namely the last two convolutional layers and the output layer, are responsible for detecting the finer details, for our case a specific tail or snout that would help the model classify the animal. Because the model kept its first layers, where as previously discussed we have the primitive features, it was able to re-learn the complex layers we replaced and obtain an accuracy of 96.60%.

Conclusion

Transfer Learning is a robust technique that helped us achieve better results than even the normal model that was trained on both datasets. As it was explained, the different models we obtained from our experiments could be further used to create an even better accuracy when it comes to the Cats and Dogs dataset. Through these experiments we also got an understanding of how each convolutional layer can affect a different part of our whole model, as well as understand what features are important to a CNN model.